

Sharmita Dey

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EDUCATION AND TRAINING

Postdoctoral researcher

ETH Zurich, Institute of Robotics and Intelligent Systems (IRIS) [11/2024 – Current]

Leading AI for assistive robotics and human-machine interaction projects

- Defining research and technical direction for a research team
- Leading the development of **novel foundation models for embodied AI**
- Lead the development of multimodal foundation models from wearable sensor data (EEG, Kinematics, Kinetics, Force) for **assistive robotics**
- Lead human-in-loop design of ML-based assistive robots, strategizing data collection pipelines
- Supervising PhD and master students
- Wrote grants on **physical AI**, AI for wearables, authored research papers

Guest Scientist

University Medical Center Göttingen / University of Göttingen, Germany [11/2024 – Current]

Leading AI for intelligent prostheses projects

Postdoctoral researcher

University Medical Center Göttingen / University of Göttingen, Germany [07/2024 – 11/2024]

Led foundation modeling and deep-learning projects for biosignals and human-machine interaction.

- Developed human-motion **foundation and diffusion models**, reprogrammed them using imitation learning
- Developed **world-models for human-robot interaction**
- Conducted hardware experiments, supervised students.

Postdoctoral researcher

Department of Computer Science, University of Göttingen, Germany [01/2023 – 07/2024]

- Developed transformer-based architectures for learning object dynamics from video, surveyed label-efficient learning (self-/semi-supervised, active and weakly supervised learning)
- Contributed to self-supervised foundation models.
- Taught seminars on self-supervised learning and co-authored research papers.

PhD in Computer Science, Faculty of Mathematics and Computer Science

University of Göttingen, Germany [2018 – 2023]

Final grade: Summa Cum Laude (highest distinction) | **Thesis:** Learning-based biomimetic strategies for developing control schemes for rehabilitation robotics

Supervisors: Prof. Dario Farina, Prof. Stephan Huckemann, and Prof. Arndt F. Schilling.

- Conducted experiments to collect multimodal time-series data (optical motion capture, IMU, EMG)
- developed multitask and continual learning **world models for assistive robots**
- executed hardware experiments with an ankle exoskeleton
- authored several publications and filed patents.

Robotics Research Affiliate

NASA Jet Propulsion Laboratory, USA [05/2021 – 12/2022]

Large-scale autonomous mapping and off-road navigation.

Developed traversability **models for quadruped robots**, applied monocular depth estimation in perceptually degraded environments, detected methane plumes in multispectral images, built traversability classifiers from Lidar point clouds and proprioceptive data, and predicted traversable paths using autoencoders.

Research Assistant

German Aerospace Center (DLR), Institute of Robotics & Mechatronics, Germany [04/2017 – 06/2018]

- Implemented domain-adaptive intent recognition from surface EMG signals for **robot arm control**
- Applied transfer learning for robust command generation
- Utilized clustering for noise filtering
- Developed an EMG-controlled simulation framework for goal-reaching and tracking tasks.

M.Sc. in Computational Logic, Institute of Artificial Intelligence, Faculty of Computer Science

Technische Universität Dresden, Germany [2015 – 2018]

Thesis: Generalized decoding of control commands from surface electromyography (sEMG) signals for a daily assistant robotic arm (EDAN).

- Achieved the highest grade in Machine Learning.

SELECTED PUBLICATIONS AND PATENTS

Full list at Google scholar:

https://scholar.google.com/citations?view_op=list_works&hl=en&hl=en&user=dRn_bawAAAAJ&pagesize=80

1. **Dey, Sharmita.**, Riener Robert, Dosen Strahinja, Albrecht Stefano. Robot Foundation Models Must Learn With us, Not Just for Us. *Science Robotics*, 2026. URL: <https://www.science.org/eprint/HUHAXBMNXAYHJPAC3YGX/full?activationRedirect=/doi/full/10.1126/scirobotics.aea1822>
2. **Dey, S.**, & Ravindran Nair, S. (2024). ReMAP: Neural Model Reprogramming with Network Inversion and Retrieval-Augmented Mapping for Adaptive Motion Forecasting. *Advances in Neural Information Processing Systems*, 37, 2519525227. URL: https://proceedings.neurips.cc/paper_files/paper/2024/file/2cdf71a65e95bf1874212f6e604c64dbPaperConference.pdf (Open Access)
3. **Dey, S.**, Paaßen, B., Nair, S. R., Boughorbel, S., & Schilling, A. F. (2025) Learning from Simulated Interactions via Multitask Prospective Rehearsal for Bionic Limb Behavior Modelling. *Transactions on Machine Learning Research*. DOI: <https://doi.org/10.48550/arXiv.2405.01114> (Open Access)
4. Vogg, R., Lüddecke, T., Henrich, J., **Dey, S.**, Nuske, M., Hassler, V., ... & Ecker, A. S. (2025). Computer vision for primate behavior analysis in the wild. *Nature Methods*, 1-13. DOI: <https://doi.org/10.1038/s41592-025-02653-y>
5. **Dey, S.**, & Schilling, A. F. (2022). A function approximator model for robust online foot angle trajectory prediction using a single IMU sensor: Implication for controlling active prosthetic feet. *IEEE Transactions on Industrial Informatics*, 19(2), 1467-1475. DOI: <https://doi.org/10.1109/TII.2022.3158935> (Open Access)
6. **Dey, S.**, Yoshida, T., & Schilling, A. F. (2020). Feasibility of training a random forest model with incomplete user-specific data for devising a control strategy for active biomimetic ankle. *Frontiers in Bioengineering and Biotechnology*, 8, 855. DOI: <https://doi.org/10.3389/fbioe.2020.00855> (Open Access)
7. **Dey, S.**, Yoshida, T., Foerster, R. H., Ernst, M., Schmalz, T., & Schilling, A. F. (2020). Continuous prediction of joint angular positions and moments: A potential control strategy for active knee-ankle prostheses. *IEEE Transactions on Medical Robotics and Bionics*, 2(3), 347-355. DOI: <https://doi.org/10.1109/TMRB.2020.3011841>
8. **Dey, S.** et.al “PrePARE: Predictive Proprioception for Agile Failure Event Detection in Robotic Exploration of Extreme Terrains”, 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). DOI: [10.1109/IROS47612.2022.9981660](https://doi.org/10.1109/IROS47612.2022.9981660)

9. Dixit, A., Fan, D. D., Otsu, K., **Dey, S.**, Agha-Mohammadi, A. A., & Burdick, J. (2024). Step: Stochastic traversability evaluation and planning for risk-aware navigation; results from the darpa subterranean challenge. *Field Robotics*, 4, 182-210. DOI: <https://doi.org/10.55417/fr.2024006>
10. **Dey, S.**, Fan, D., Schmid, R., Dixit, A., Otsu, K., Touma, T., ... & Agha-Mohammadi, A. A. (2022, October). Prepare: Predictive proprioception for agile failure event detection in robotic exploration of extreme terrains. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 4338-4343). IEEE. DOI: <https://doi.org/10.1109/IROS47612.2022.9981660>
11. Schmid, R., Atha, D., Schöller, F., **Dey, S.**, Fakoorian, S., Otsu, K., ... & Agha-Mohammadi, A. A. (2022, October). Self-supervised traversability prediction by learning to reconstruct safe terrain. In *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 12419-12425). IEEE. DOI: <https://doi.org/10.1109/IROS47612.2022.9981368>
12. Morrell, B., Otsu, K., Agha, A., Fan, D. D., Kim, S. K., Ginting, M. F., ..., **Dey, S.**, ... & Burdick, J. (2024). An addendum to NeBula: toward extending team CoSTAR's solution to larger scale environments. *IEEE Transactions on Field Robotics*, 1, 476-526. DOI: [10.1109/TFR.2024.3430891](https://doi.org/10.1109/TFR.2024.3430891)
13. **Dey, S.**, De Schultz, N., & Schilling, A. F. (2023, September). Why Hard Code the Bionic Limbs When They Can Learn From Humans? In *2023 International Conference on Rehabilitation Robotics (ICORR)* (pp. 1-6). IEEE. DOI: <https://doi.org/10.1109/ICORR58425.2023.10304817>
14. **Dey, S.**, Paaßen, B., Ravindran Nair, S., Boughorbel, S., & Schilling, A. F. (2024). Continual Imitation Learning for Prosthetic Limbs. *arXiv e-prints*, arXiv:2405. DOI: <https://doi.org/10.48550/arXiv.2405.01114> (Open Access)
15. **Dey, S.**, & Schilling, A. F. (2024). Embracing Large Language and Multimodal Models for Prosthetic Technologies. *arXiv preprint arXiv:2403.04974*. DOI: <https://doi.org/10.48550/arXiv.2403.04974> (Open Access)
16. **Dey, S.**, Yoshida, T., Foerster, R. H., Ernst, M., Schmalz, T., Carnier, R. M., & Schilling, A. F. (2021). A hybrid approach for dynamically training a torque prediction model for devising a human-machine interface control strategy. *arXiv preprint arXiv:2110.03085*. DOI: <https://doi.org/10.48550/arXiv.2110.03085> (Open Access)
17. **Dey, S.**, Boughorbel, S., & Schilling, A. F. "Learning a Shared Model for Motorized Prosthetic Joints to Predict Ankle-Joint Motion." *NeurIPS 2021 Workshop and Machine Learning for Health (Spotlight)*. DOI: <https://doi.org/10.48550/arXiv.2111.07419> (Open Access)
18. **Dey, S.** and Nair, S. "Enhancing Joint Motion Prediction for Individuals with Limb Loss Through Model Reprogramming". 2024 International Conference on Learning Representations (ICLR) Workshops. DOI: <https://doi.org/10.48550/arXiv.2403.06569>
19. Patent Filing WO2021005006 — Methods and Systems for Controlling Active Prostheses, **S. Dey**, M. Eslamy, A. Schilling. URL: <https://patents.google.com/patent/WO2021005006A2/en>
20. Patent Filing EP4372617A1 — Method and Apparatus for Training a Machine-Learning Model, **S. Dey**, A. Schilling. URL: <https://patents.google.com/patent/EP4372617A1/en>
21. **Dey, S.** (2025). Redefining Robot Generalization Through Interactive Intelligence. *arXiv preprint arXiv:2502.05963*. DOI: <https://doi.org/10.48550/arXiv.2502.05963> (Open Access)
22. **Dey, S.**, & Nair, S. R. (2025). Cross-Modal Diffusion for Biomechanical Dynamical Systems Through Local Manifold Alignment. *arXiv preprint arXiv:2503.12214*. DOI: <https://doi.org/10.48550/arXiv.2503.12214> (Open Access)
23. **Dey, S.**, & Paez-Granados, D. PathoFM: Toward a Foundation Model for Pathological Gait. In *NeurIPS 2025 Workshop on Learning from Time Series for Health*. URL: <https://openreview.net/forum?id=Hq9Afq7EMk>
24. Lee, J., Mauceri, S., **Dey, S.**, Rao, A.A., Alimo, R., Thorpe, A., Jongaramrungruang, S. and Duren, R., 2021, December. Methane Plume Detection with Future Orbital Imaging Spectrometers. In *AGU Fall Meeting Abstracts* (Vol. 2021, pp. GC15B-0690). URL: <https://essopenarchive.org/doi/full/10.1002/essoar.10509499.1>
25. **Dey, S.** Continual world model with counterfactual simulated interactions. In *Women in Machine Learning Workshop@ NeurIPS 2025*. URL: <https://openreview.net/forum?id=prO4dG2QAI>
26. **Dey, S.**, Yoshida, T., Ernst, M., Schmalz, T. and Schilling, A.F., 2019, September. A random forest approach for continuous prediction of joint angles and moments during walking: An implication for controlling active knee-ankle prostheses/orthoses. In *2019 IEEE International conference on Cyborg and bionic systems (CBS)* (pp. 66-71). IEEE. DOI: [10.1109/CBS46900.2019.9114439](https://doi.org/10.1109/CBS46900.2019.9114439)

27. D D. Fan, **S Dey**, A. Mohammadi, and E.Theodorou, "Learning Risk-aware Cost maps for Traversability in Challenging Environments", 2022 IEEE International Conference on Robotics and Automation (ICRA). DOI: [10.1109/LRA.2021.3125047](https://doi.org/10.1109/LRA.2021.3125047)
28. **S Dey** et. al. "A support vector regression approach for continuous prediction of ankle angle and moment during walking: An implication for developing a control strategy for active ankle." IEEE International Conference on Rehabilitation Robotics (ICORR): Vol. 2019. DOI: [10.1109/ICORR.2019.8779445](https://doi.org/10.1109/ICORR.2019.8779445)
29. **S Dey** et. al. "Data-driven gait-predictive model for anticipatory prosthesis control" IEEE International Conference on Rehabilitation Robotics (ICORR): 2022. DOI: [10.1109/ICORR55369.2022.9896505](https://doi.org/10.1109/ICORR55369.2022.9896505)
30. **Dey, S.**, Yoshida, T. and Schilling, A.F., 2020. A Machine Learning Based Strategy for Predicting the Sagittal Angles in real-time for Actuating an Active Knee-Ankle Prosthesis. *Zeitschrift für Orthopädie und Unfallchirurgie*, 158(S 01), pp.DKOU20-534. DOI: <https://doi.org/10.1055/s-0040-1717439>

TEACHING AND SUPERVISION

University teaching

- **Introduction to Assistive Robotics** — Winter semesters 2019/20, 2020/21 and 2021/22.
- **Machine Learning and Computational Neuroscience Seminar** — Summer semesters 2023 and 2024.
- **Current Topics in Deep Learning** — Winter semester 2023/24.
- **Introduction to Graph Transformers** — Winter semester 2023/24.

Guest lectures

- **Intelligent Human Machine Interactions** University Medical Center, Goettingen – 2023.
- **Intelligent Human centred Robotics** Operative medicine module, Life Science Factory, Goettingen – 2021 and 2022.
- **Intelligent Control of Assistive Robotics** German Congress of Orthopaedics and Traumatology, 2023

Selected student supervision

- **Laurin Wick** - Master project supervision, ETH Zurich, 2026
- **Christian Popescu** - Master project supervision, ETH Zurich, 2026
- **Leander Reinberger** – Master project supervision, ETH Zurich, 2025.
- **Francesco Pavia** – Master thesis supervision, ETH Zurich, 2025.
- **Heba Hossein** – Master project supervision, ETH Zurich, 2025.
- **Nuray Agaoglu** – Master thesis supervision, ETH Zurich, 2025.
- **Gloria Edumamba** – Master thesis co- supervision, ETH Zurich, 2025.
- **Niklas De Schultz** – Bachelor thesis supervision, University of Goettingen 2022.
- **Georg Eckardt** – Master thesis supervision, University of Goettingen 2023.
- **Tom Dieck** – Master project supervision, University of Goettingen 2020.
- **Antonia Bernecker** – Master project supervision, University of Goettingen 2020.

HONOURS AND AWARDS

- [06/2024] **Cordi Fellowship** from INRIA France - Awarded fellowship to pursue independent research
- [05/2024] **Scientific Startup Idea Award for intelligent assistive limbs** at the Lift-off Start-up Competition.
- [02/2024] **Invited talk** at Max Planck Institute for Intelligent Systems, Stuttgart
- [01/2024] **Selection to GWG incubator program** Startup idea on intelligent assistive limbs.
- [11/2023] **Invited talk** at German Congress of Orthopaedics and Traumatology
- [02/2023] **Summa Cum Laude** highest distinction for doctoral work at University of Goettingen
- [05/2021] **Selected for highly competitive NASA JPL Visiting Researcher's Program** acceptance rate ~7 % [2021] **Finalist in DARPA Subterranean Robotics Challenge** Team: CoSTAR

[2021] **Spotlight Panel Talk** NeurIPS 2021 Workshop (Machine Learning for Health)

[2019] **Finalist for Best Conference Paper** IEEE CBS [2017] **Dr. Kanako Miura Travel Grant Award**, IEEE Humanoids 2017

GRANT SUBMISSIONS

Received

- **Heilbronn Innovation Artificial Intelligence Grant**
- **Women In Machine Learning**

Under consideration

- **ERC Starting Grant 2026**
- **Volkswagen Stiftung Grant** — Principal Investigator. Proposal on “*Responsible AI for Wearable Systems*” with a budget of 2 000 000 EUR.

SERVICE AND OUTREACH

Mentoring and volunteering

Mentor for Machine Learning for Health (ML4H) and Women in Machine Learning (WiML) programmes, supporting early-career researchers and under-represented groups in machine learning and health.

Reviewer activities

Reviewer for leading conferences and journals, including Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), International Conference on Machine Learning (ICML), Artificial Intelligence and Statistics (AISTATS), IEEE International Conference on Robotics and Automation (ICRA), Robotics and Automation Letters (RAL), IEEE International Conference on Intelligent Robots and Systems (IROS), IEEE/ASME Transactions on Mechatronics, IEEE Transactions on Biomedical Engineering, IEEE Journal of Biomedical and Health Informatics, IEEE Transactions on Neural Systems and Rehabilitation Engineering, IEEE International Conference on Rehabilitation Robotics (ICORR), and IEEE Transactions on Industrial Informatics.

RELEVANT SKILLS

Programming: Python, MATLAB | **Deep learning frameworks:** PyTorch, TensorFlow | **Machine learning libraries:** scikit-learn, NumPy, SciPy | **Sensor technologies:** IMU, EMG, force-torque | **Robotics:** robot operating systems (ROS), hardware prototyping | **AI methods:** supervised and self-supervised learning, foundation models, continual and multitask learning, generative models, and world models, reinforcement learning | **Modeling and Simulation:** Opensim, MSC Adam, Vicon